

CNN-based Gripping-force Estimation Using Optical Muscle Deformation Sensors

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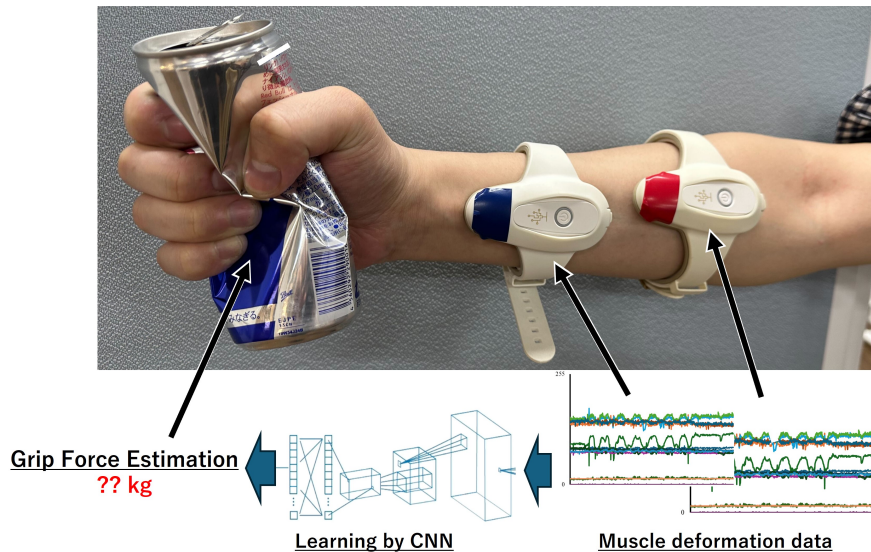


Figure 1: Grip force estimation using muscle deformation data from optical sensors.

Abstract

Hand-gripping strength is vital in physical activity but is challenging to quantify from arm appearance. This study aimed to estimate grip strength using forearm muscle activity, addressing challenges like sensor attachment position and muscle-skin spatial relationships. A deep-learning method with optical muscle deformation sensors was developed to mitigate these factors. Two-line circular-array sensing collected data during 5–15 kg grip strength from 10 participants across five arm orientations, which were combined into a single dataset. This dataset trained convolutional neural networks (CNN) and support vector regression (SVR). Evaluation showed that while SVR failed to estimate grip strength under posture changes, CNN achieved ± 1 kg accuracy and generalized across individuals without calibration. These findings demonstrate CNN's robustness and significant improvement over conventional methods, paving the way for practical applications of optical muscle deformation systems in various fields. However, adaptation to sweat-induced impedance changes remains a future challenge.

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1 Introduction

1.1 Background

The hand plays a vital role in physical interactions, with hand-gripping being fundamental to human activity. Grip strength is crucial in sports like golf, tennis, hockey, and baseball [4], requiring either power or precision grips [28]. Studies show elite athletes have greater grip strength than their less successful peers [24]. Additionally, grip strength serves as a health index in aging populations, correlating with better health outcomes [24].

Grip force is essential not only for holding objects but also for perceiving weight. While humans learn handling techniques by imitation, the exact force applied by fingers remains unquantifiable through observation. Quantifying grip and finger forces aids in object handling, supports imitation learning for robots, and enhances interactions in virtual and augmented reality [13].

Wearable devices enable precise measurements of grip strength, finger pressure, and force distribution, offering insights for sports

performance, rehabilitation, and health monitoring [1]. These devices also detect muscle strength or motor control issues early, turning subjective observations into data-driven insights. Forearm muscles significantly influence grip strength, with postures like pronation or supination causing variations in muscle strain and deformation [3]. Sweating during activity further alters biosignal readings, necessitating frequent calibration [7]. Although calibration games [7] make this process engaging, they require special devices and time. Consequently, an estimation system that is resilient to changes in the posture of an arm and sweating without calibration is essential.

1.2 Related Works

Wearable sensors are widely used to estimate hand-gripping or finger-pushing force, but all methods require a calibration process. Calibration involves creating training data by pairing sensor values with correct force values [23]. Biosignals used for estimation include electromyography (EMG) [17], bone-conducted sound [8], skin deformation [21], and muscle deformation [20]. Calibration is necessary whenever sensor placement, body posture, or skin conditions (e.g., sweating) change.

Fingertip-mounted wearable sensors directly capture biosignals related to finger-pushing force. Saito et al. [21] used a photo sensor to measure skin deformation, while Kristanto et al. [14] applied Hall-effect tactile sensors. Funato et al. [8] estimated finger-pushing force using bone-conducted sound generated by an actuator. Liu et al. [15] developed stretchable electric skins for hand gloves, enabling force measurement and haptic feedback [22].

Forearm-mounted sensors avoid interference between the hand and objects during force estimation. EMG, widely used for measuring muscle activity, represents action potentials during contractions. Hou et al. [11] and Hoozemans [10] used EMG to estimate gripping force, employing methods like root-mean-square calculations [11] and generalized regression neural networks [25]. Martinez [16] developed an online method for amputees, while Zhang et al. [29] applied EMG for human-robot collaboration. Other approaches include inverse EMG imaging [6] and dynamic force reconstruction using deep learning [27]. However, EMG signals are affected by skin conditions like sweating and sensor displacement.

Many studies have explored gesture detection by placing sensors on the forearm [26], [5]. However, these studies mainly focus on gestures involving large finger movements and are not applicable to estimating fingertip force transitions. Additionally, they do not verify the estimation of force exerted by the fingers in the absence of visible finger movements.

Muscle deformation in the forearm can estimate grip force without interfering with hand-object interaction. Funato et al. [9] used bone-conducted sound sensing, while Cho et al. [2] and Kato et al. [12] developed distance sensor systems to measure deformation. Hosono et al. [20] combined muscle deformation sensors with a photo sensor array to measure fingertip force. These methods avoid some limitations of EMG but still require calibration.

Despite their potential, existing methods for estimating grip or finger force involve tedious calibration processes and specialized equipment [7], limiting their practicality for daily use.

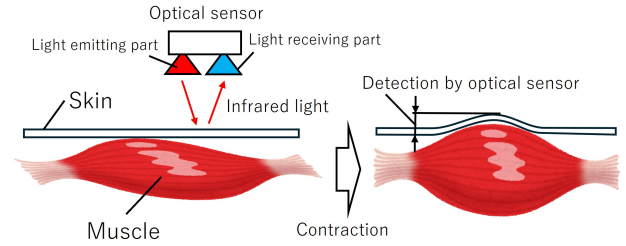


Figure 2: Muscle deformation measurement using optical sensors. The optical sensor consists of a light emitter and a light receiver. The emitter emits infrared light towards the skin, and the receiver measures the reflected light. Therefore, the optical sensor can detect the raised portion of the skin pushed up by the muscles.

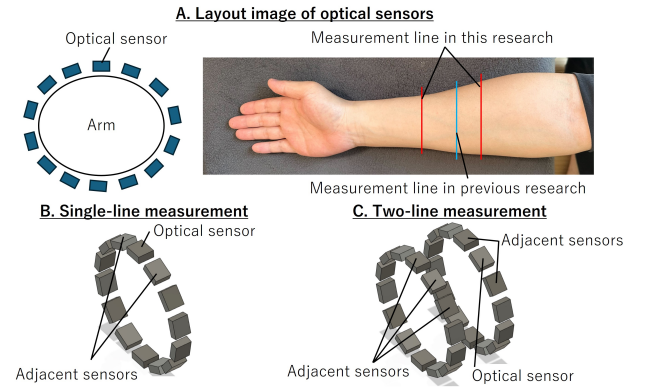


Figure 3: Arrangement of the optical sensors of this system. Multiple optical sensors are arranged to surround the forearm, measuring muscle deformation. In conventional systems, optical sensors surrounding the forearm were positioned in a single rotation. However, in this system, by employing two full rotations of optical sensors, we aim to construct a grip force estimation system unaffected by supination and pronation.

1.3 Objective

Muscle measurement is affected by sweat (skin impedance changes), sensor attachment position, and the spatial relationship between muscles and skin. This study aims to develop a deep-learning-based method for estimating grip force using 2D muscle deformation to address these variations. We hypothesized that measuring in two dimensions enables the analysis of relationships between adjacent muscles or areas within the same muscle, enhancing robustness. To counteract skin impedance changes, we applied infrared sensors to measure muscle deformation. By placing multiple infrared sensors around the forearm, the system aims to estimate grip force even during hand pronation or supination.

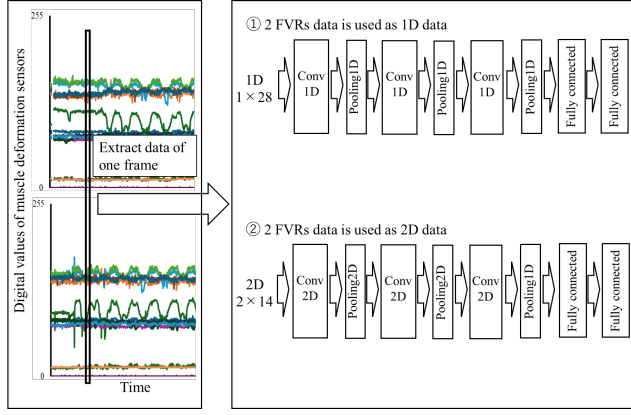


Figure 4: CNN model architecture estimating hand-gripping force from muscle deformation data. There were two types of developed CNN estimators; muscle deformation data were processed as 1D and 2D data.

2 System

A system is designed to estimate grip force during pronation or supination. Multiple infrared sensors, consisting of emitters and receivers, are placed around the forearm to measure muscle deformation. Biological tissue strongly scatters near-infrared light [19]. By emitting infrared light toward the skin and measuring the reflected light from both the skin and deeper muscles, these sensors indirectly capture muscle deformation during contraction (Figure 2). The positional relationship between muscles and skin is crucial for accurate measurement, but pronation and supination cause significant changes in the arrangement of the radius, ulna, and muscles, complicating this relationship. Traditional systems with a single circuit of sensors struggle to compensate for these changes, making grip force estimation difficult (Figure 3 B).

We hypothesized that gripping-related muscle deformation appears consistently in longitudinal and circumference patterns, even during pronation or supination. To address this, circular-array sensors were placed at the wrist and elbow sides (Figure 3 C). Conventional methods typically use a single sensor line, producing $1 \times N$ vectors (N : number of sensors around the circumference), with each sensor adjacent to two others. The proposed system introduces a second dimension along the arm's length, creating $2 \times N$ vectors with five adjacent sensors, enabling image-like processing.

To detect spatial patterns across wrist and elbow sensor arrays, we employed a convolutional neural network (CNN), which is equivariant to translation, to adaptively extract features. Figure 4 illustrates the data acquisition and CNN workflow. Muscle deformation data from two infrared sensors are input into the CNN. Both 1D and 2D data processes (Table 1 and 2) were tested. The CNN used ReLU activation functions, convolutional layers with kernel sizes of 3, and dropout layers to prevent overfitting. Fully connected layers output regression results. Increasing filters in convolutional layers allowed the model to learn detailed features, capturing specific patterns across channels. Training included 15,000 epochs.

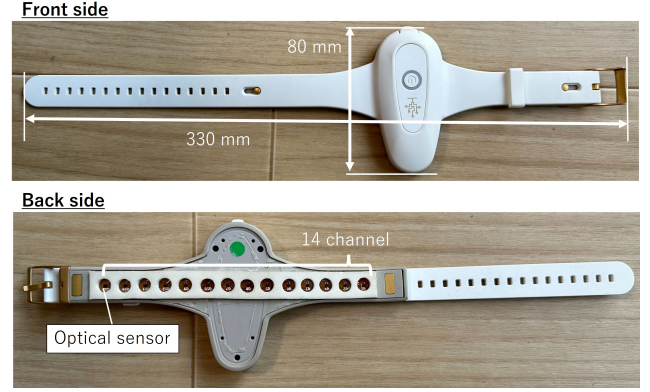


Figure 5: Description of FirstVR: The FirstVR is a band-shaped wearable device equipped with 14-channel optical sensors on the backside of the band. By wrapping it around, the optical sensors can be positioned on the forearm. Its dimensions are 80mm in height, 330mm in width, and it weighs 56g.

Table 1: Constitutions of CNN1D algorithm

No.	Layer Type	Parameters
1st	Conv1D	Filters: 32, Kernel size: 3, Padding: same, Activation: ReLU
2nd	MaxPooling1D	Pool size: 2, Stride: 1, Padding: same
3rd	Conv1D	Filters: 64, Kernel size: 3, Padding: same, Activation: ReLU
4th	MaxPooling1D	Pool size: 2, Stride: 1, Padding: same
5th	Conv1D	Filters: 128, Kernel size: 3, Padding: same, Activation: ReLU
6th	MaxPooling1D	Pool size: 2, Stride: 1, Padding: same
7th	Dropout	Drop rate: 0.5
8th	Dense	Units: 10, Activation: ReLU
9th	Flatten	-
10th	Dense	Units: 1, Activation: Linear

Table 2: Constitutions of 2D CNN algorithm

No.	Layer Type	Parameters
1st	Conv2D	Filters: 32, Kernel size: (3, 3), Padding: same, Activation: ReLU
2nd	MaxPooling2D	Pool size: 2, Stride: 1, Padding: same
3rd	Conv2D	Filters: 64, Kernel size: (3, 3), Padding: same, Activation: ReLU
4th	MaxPooling2D	Pool size: 2, Stride: 1, Padding: same
5th	Conv2D	Filters: 128, Kernel size: (3, 3), Padding: same, Activation: ReLU
6th	MaxPooling2D	Pool size: 2, Stride: 1, Padding: same
7th	Dropout	Drop rate: 0.5
8th	Dense	Units: 10, Activation: ReLU
9th	Flatten	-
10th	Dense	Units: 1, Activation: ReLU

3 Materials and Methods

3.1 Participants

Ten participants took part in the study, comprising 10 males, with the age range from 22 to 32 years. All participants were provided with an explanation of the experimental procedures and gave informed consent before the commencement of the study. Additionally, participants were briefed on the procedure for interrupting the experiment at any time if needed. However, the specific objectives of the experiment were not disclosed to the participants. This experiment was approved by The Japanese Society for Wellbeing Science and Assistive Technology (ID:22-13).

3.2 Experimental Setup

In this research, we utilized two FirstVRs (H2L Co., Ltd.) as optical sensors attached to the forearm. Figure 5 illustrates the front and back views of the FirstVR. On the backside of the FirstVR, 14-channel optical sensors are arranged in a single row, with each sensor equipped with one emitter and one receiver. The size of FirstVR is 80mm × 330mm, and it weighs 56g, making it a wearable device that does not impose any burden on humans when placed on the forearm. By emitting infrared light toward the human body and measuring the intensity of the reflected light with the receiver, the optical sensors can measure muscle deformation. Additionally, by wrapping the FirstVR's band with the optical sensors around the arm, it is possible to position the sensors around the outer circumference of the forearm.

The experimental setup employed in this research consisted of a grip strength dynamometer (TKK5401, Takei Scientific Instruments Co., Ltd.), two FirstVRs (H2L Co., Ltd.), and a PC. The FirstVRs transmit the 14-channel optical sensor data, to the PC using Bluetooth Low Energy. The data transmission frequency is set at 10 Hz.

3.3 Dataset

The data collection process is shown in Figure 6. Subjects wore two FirstVR devices on the forearm of their right hand (Figure 6 A) and used a dynamometer to measure grip strength. Muscle deformation data were collected using optical sensors while adjusting grip strength (5–15kg in 1kg increments, 11 conditions) and hand posture. Hand posture included five specific positions: palm facing upward, downward, rightward, leftward, and twisting from top to bottom (Figure 6 B and C). This resulted in 55 conditions per subject, with data recorded for 5 seconds per condition. A tolerance of ±0.5kg was allowed for grip strength, recorded to the first decimal place. Subjects reattached the FirstVR device at least twice during the session. All data were anonymized and consolidated into a single dataset.

3.4 Experimental Procedure

Using the collected dataset, grip strength was estimated with CNN and SVR. The dataset was split into 75% for training and 25% for evaluation, with parameters adjusted to minimize RMSE. Post-learning, MAE was calculated for evaluation, as shown in Equations 1 and 2, respectively.

Table 3: MAE, RMSE, and coefficient of determinations for each experimental condition.

Model	FVR	Data	MSE	RMSE	R2
CNN	2	1D	0.85	1.36	0.83
CNN	2	2D	1.01	1.55	0.76
CNN	1	1D	1.54	2.33	0.47
SVR	1	1D	2.54	2.93	0.12
SVR	2	1D	2.13	2.59	0.32

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} (F_i - \hat{F}_i)^2} \quad (1)$$

$$MAE = \frac{1}{N} \sum_{i=0}^{N-1} |F_i - \hat{F}_i| \quad (2)$$

Where N is the total number of samples, F represents the measured grip force, and $\hat{F}$ denotes the estimated grip force. SVR hyperparameters were determined based on grid-search optimization, and trained using the best hyperparameters.

This section details the learning conditions. CNN learning used two FirstVR devices to create 1D and 2D datasets. The 1D dataset was a 1×28 format combining data from two devices (14 sensors each) without reordering. The 2D dataset treated the data as a 2×14 matrix, capturing forearm muscle deformation as surface data for detailed learning of the muscle deformation-grip force relationship. Additionally, we validated the system's superiority by using only the elbow-side FirstVR, with input data as a 1×14 matrix. The number of data used for training was the same under all conditions. All training sessions used 30,000 epochs.

For SVR learning, calculations were performed using two FirstVR devices and only the elbow-side device. This enabled comparison between SVR, used in previous research, and CNN.

4 Results

When conducting CNN learning with two FirstVR devices, the results are shown in Figures 7 and 8, where input data is represented as 1D (1×28) and 2D (2×14), respectively. Figures 9 and 10 show the results of CNN and SVR using only the elbow-side FirstVR, while Figure 11 shows SVR results with two devices. These figures plot actual vs. estimated grip strength on the x-axis and y-axis, respectively, and display RMSE and the coefficient of determination. Table 3 summarizes MAE and the coefficient of determination.

Comparing Figures 7 and 8, 1D input data achieves higher accuracy (higher coefficient of determination, lower MAE and RMSE) than 2D. Figures 9 vs. 7 and 10 vs. 11 indicate that using two devices improves all metrics for both CNN and SVR, with CNN achieving a coefficient of determination exceeding 0.75. Finally, Figures 9 vs. 10 and 7 vs. 11 show that CNN consistently outperforms SVR under the same input conditions.

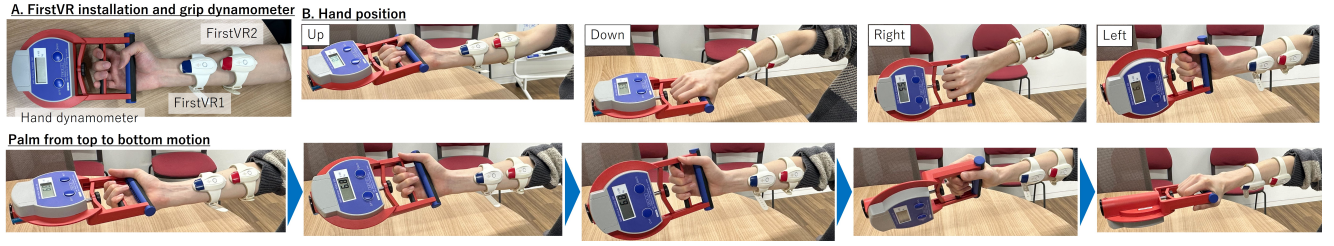


Figure 6: Description of the Experiment: A. FirstVR installation and grip dynamometer depicts the procedure during the experiment. The experiment involves placing one FirstVR device each on the wrist side and elbow side of the right forearm, while the participant grips the dynamometer with their right hand. B. Hand positions are illustrated, representing five conditions: palm facing upwards, downwards, rightwards, leftwards, and twisting from top to bottom.

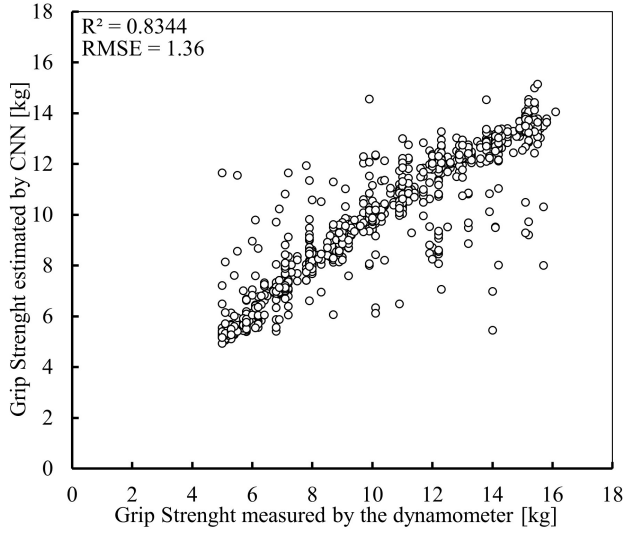


Figure 7: Relationship Between Estimated and Actual Grip Strength Values in CNN Learning with a 1D (1×28) Dataset Using Two FirstVR Devices.

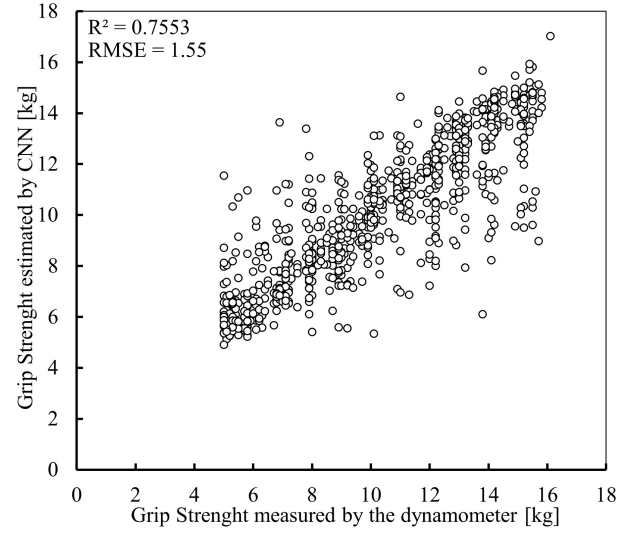


Figure 8: Relationship Between Estimated and Actual Grip Strength Values in CNN Learning with a 2D (2×14) Dataset Using Two FirstVR Devices.

5 Discussion

5.1 Grip strength estimation model by optical muscle deformation sensors

First, the estimation accuracy of CNN and SVR was compared using 1D (1×28) data from two FirstVR devices. SVR showed a low coefficient of determination (0.32), while CNN achieved 0.83 with an MAE of 0.85, corresponding to an accuracy of approximately ± 1 kg. Since these results were obtained from the same dataset, the differences in estimation accuracy are due to the variations in the algorithms. CNN's ability to extract inter-sensor relationships outperformed SVR, which processes each data point independently. A prior study achieved finger force estimation under significant changes in arm posture by combining SVR with an IMU [18]. In contrast, this study demonstrated the feasibility of force estimation under twisted conditions without using an IMU, utilizing two FirstVR devices and CNN. This was achieved because CNN's capability of translation invariance compensated for variations in

sensor readings caused by arm pronation, supination, and sensor attachment errors, leading to improved estimation accuracy and robustness.

Next, differences between 1D and 2D input data in CNN were examined. While 2D data theoretically provides richer feature extraction, 1D data achieved higher accuracy with a better coefficient of determination. Since these results were also obtained from the same dataset, the data dimensionality is considered to influence the estimation accuracy. In this study, the positional relationship between sensors in the 1D muscle deformation data remained consistent, as each FirstVR device was mechanically fixed, preventing changes due to reattachment. In contrast, the relative positions between sensors in the 2D muscle deformation data varied due to the attachment of two separate FirstVR devices. However, these positional variations were not quantified. During the experiment, each subject reattached the FirstVR devices at least twice, making the sensor positions in the 2D data more variable compared to the 1D data. As a result, despite the expectation that the 2D dataset

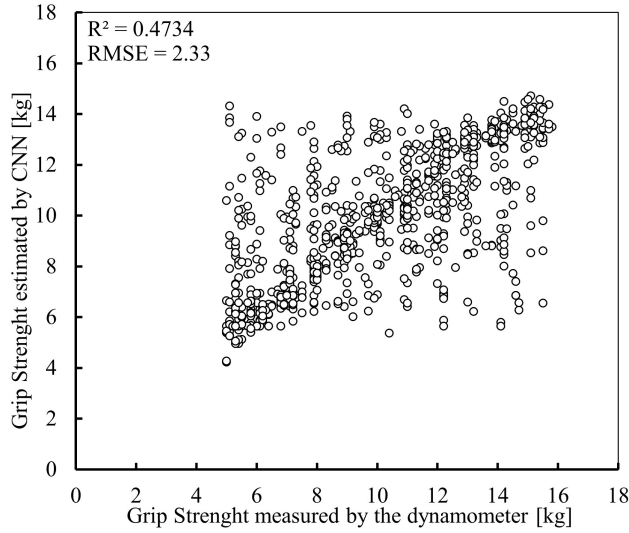


Figure 9: Relationship Between Estimated and Actual Grip Strength Values in CNN Learning with a 1×14 Dataset Using one FirstVR Devices.

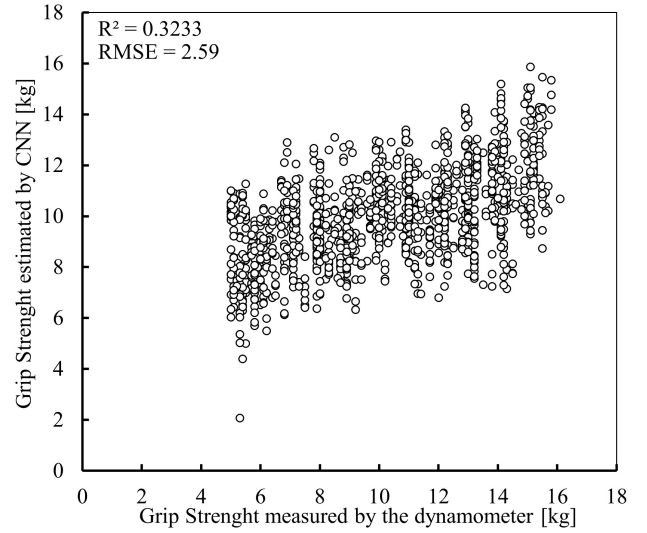


Figure 11: Relationship Between Estimated and Actual Grip Strength Values in SVR Learning with a 1×28 Dataset Using Two FirstVR Devices.

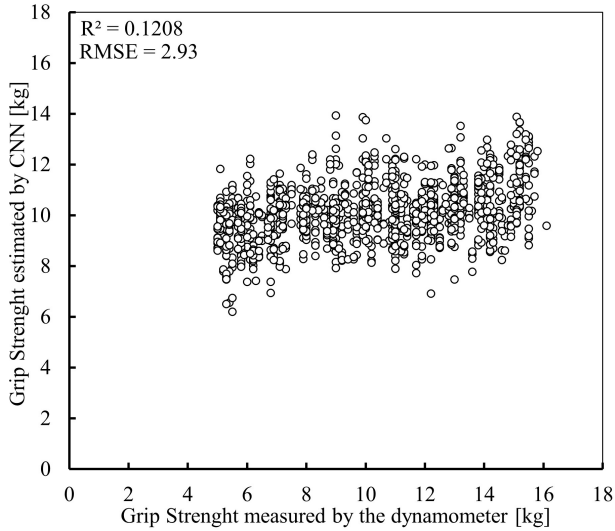


Figure 10: Relationship Between Estimated and Actual Grip Strength Values in SVR Learning with a 1×14 Dataset Using one FirstVR Devices.

would provide more inter-sensor relationships and facilitate feature extraction, the 1D dataset achieved higher estimation accuracy.

The impact of the number of FirstVR devices on estimation accuracy was also analyzed. Using two devices (28 sensors) significantly improved accuracy, with the coefficient of determination ranging from 0.76 to 0.83, compared to one device, which achieved a coefficient of determination of approximately 0.3. This improvement in accuracy is partly due to the increase in input data, but it is primarily attributed to the use of a two-row sensor arrangement

on the arm. Two devices enabled measurements at multiple positions along the forearm's twisting axis, facilitating the extraction of features related to muscle deformation and twisting.

In conclusion, combining two FirstVR devices improves grip strength estimation accuracy, achieving an error margin within ± 1 kg. CNN outperformed SVR by capturing spatial patterns and inter-sensor relationships, making it robust to variations such as arm pronation, supination, and muscle deformation. While 2D data holds potential, consistent sensor placement and optimized configurations remain key to further improving accuracy.

5.2 Grip strength estimation by optical muscle deformation sensors for motion analysis

The method of measuring muscle deformation with optical sensors resists changes in skin impedance caused by sweat and humidity, outperforming traditional methods like EMG. This resistance ensures consistent data collection even during physical activity, enhancing reliability and making it ideal for motion analysis. By eliminating the need for sensors between the hand and objects, as seen in sensor gloves, it allows natural movements and smoother everyday motion analysis. The FirstVR device further demonstrated accurate grip strength estimation despite multiple reattachments, showing tolerance to slight sensor displacement. This capability ensures stable and reliable measurements during real-world operations, reinforcing its practicality for motion analysis.

5.3 The necessity of calibration for grip strength estimation

One of the key features of this study is that data from 10 participants was used as a single dataset. In conventional studies that estimate finger force from muscle deformation or EMG, datasets

are typically divided by participant, and the model must be optimized for each individual in order to estimate force accurately. However, in this study, we demonstrated that muscle deformation data could be converted into grip strength using CNN, even when combining data from multiple participants into a single dataset. This suggests that there are common features across all participants. Unlike traditional methods, this approach has the potential to lead to the development of a highly versatile model that does not require individual adjustments for each participant. By capturing these common features, it is possible to construct a grip strength estimation method that can be applied to a wide range of individuals without the need for personalized learning. Furthermore, this method indicates the potential to eliminate the need for individual calibration, saving both time and effort, which contributes to its practicality for real-world applications.

5.4 Limitation

The results of this study suggest that changes in sensor position negatively impact grip strength estimation when using optical sensors for muscle deformation measurement. However, no quantitative evaluation of these positional shifts was conducted. During the experiment, FirstVR devices were visually positioned in the same location by the experimenters. Although sensor positions changed upon reattachment, no significant displacement was observed. Therefore, in CNN-based estimation using 2D data, even minor sensor position shifts may have contributed to reduced estimation accuracy.

Additionally, this study did not evaluate the processing time for estimation. Since the CNN-based algorithm used in this study has a relatively simple structure, significant delays in estimation are unlikely. However, when using low-performance computing devices, delays may occur. Considering the overall system, computational cost may become a more significant factor than the cost of the FirstVR itself when implementing this approach.

6 Conclusions

In this study, a system for estimating grip strength was developed and evaluated using optical muscle deformation sensors placed on the forearm. The results showed that grip strength could be estimated with an accuracy of ± 1 kg, even during forearm pronation and supination, by wrapping the sensors around the forearm twice and training the model with CNN. CNN significantly outperformed SVR in estimation accuracy, demonstrating its superiority.

A key finding was that CNN maintained high accuracy even after repeated detachment and reattachment of the muscle deformation sensors during experiments. The system also performed well with datasets combining data from multiple subjects, suggesting CNN's ability to extract common features across individuals without requiring individual calibration. CNN's capacity to learn relationships between adjacent sensor values, similar to image processing, likely contributes to its robustness. These results highlight the potential for practical applications of optical muscle deformation sensors. Unlike traditional methods like SVM and SVR, which struggle with muscle structure changes, CNN, a translation-equivariant feature extractor, can flexibly adapt, opening opportunities for further research and applications.

Future work will focus on improving grip strength estimation accuracy by acquiring smoother data and developing wearable devices capable of measuring 2D muscle deformation to reduce the impact of sensor placement variability. Expanding the dataset to include more diverse participants will help create a more robust and generalized model. Additionally, combining the CNN-based system with filtering functions and integrating an IMU will be explored for real-time, high-accuracy grip force estimation.

Acknowledgments

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